Sintilimab plus a bevacizumab biosimilar (IBI305) versus sorafenib in unresectable hepatocellular carcinoma (ORIENT-32): a randomised, open-label, phase 2–3 study

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Summary

Background China has a high burden of hepatocellular carcinoma, and hepatitis B virus (HBV) infection is the main causative factor. Patients with hepatocellular carcinoma have a poor prognosis and a substantial unmet clinical need. The phase 2–3 ORIENT-32 study aimed to assess sintilimab (a PD-1 inhibitor) plus IBI305, a bevacizumab biosimilar, versus sorafenib as a first-line treatment for unresectable HBV-associated hepatocellular carcinoma.

Methods This randomised, open-label, phase 2–3 study was done at 50 clinical sites in China. Patients aged 18 years or older with histologically or cytologically diagnosed or clinically confirmed unresectable or metastatic hepatocellular carcinoma, no previous systemic treatment, and a baseline Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 were eligible for inclusion. In the phase 2 part of the study, patients received intravenous sintilimab (200 mg every 3 weeks) plus intravenous IBI305 (15 mg/kg every 3 weeks). In the phase 3 part, patients were randomly assigned (2:1) to receive either sintilimab plus IBI305 (sintilimab–bevacizumab biosimilar group) or sorafenib (400 mg orally twice daily; sorafenib group), until disease progression or unacceptable toxicity. Randomisation was done using permuted block randomisation, with a block size of six, via an interactive web response system, and stratified by macrovascular invasion or extrahepatic metastasis, baseline α-fetoprotein, and ECOG performance status. The primary endpoint of the phase 2 part of the study was safety, assessed in all patients who received at least one dose of study drug. The co-primary endpoints of the phase 3 part of the study were overall survival and independent radiological review committee (IRRC)-assessed progression-free survival according to Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1 in the intention-to-treat population. The study is registered with ClinicalTrials.gov, NCT03794440. The study is closed to new participants and follow-up is ongoing for long-term outcomes.

Findings Between Feb 11, 2019 and Jan 15, 2020, we enrolled 595 patients: 24 were enrolled directly into the phase 2 safety run-in and 571 were randomly assigned to sintilimab–bevacizumab biosimilar (n=380) or sorafenib (n=191). In the phase 2 part of the trial, 24 patients received at least one dose of the study drug, with an objective response rate of 25·0% (95% CI 9·8–46·7). Based on the preliminary safety and activity data of the phase 2 part, in which grade 3 or worse treatment-related adverse events occurred in seven (29%) of 24 patients, the randomised phase 3 part was started. At data cutoff (Aug 15, 2020), the median follow-up was 10·0 months (IQR 8·5–11·7) in the sintilimab–bevacizumab biosimilar group and 10·0 months (8·4–11·7) in the sorafenib group. Patients in the sintilimab–bevacizumab biosimilar group had a significantly longer IRRC-assessed median progression-free survival (4·6 months [95% CI 4·1–5·7]) than did patients in the sorafenib group (2·8 months [2·7–3·2]; stratified hazard ratio [HR] 0·56, 95% CI 0·46–0·70; p<0·0001). In the first interim analysis of overall survival, sintilimab–bevacizumab biosimilar showed a significantly longer overall survival than did sorafenib (median not reached [95% CI not reached–not reached] *vs* 10·4 months [8·5–not reached]; HR 0·57, 95% CI 0·43–0·75; p<0·0001). The most common grade 3–4 treatment-emergent adverse events were hypertension (55 [14%] of 380 patients in the sintilimab–bevacizumab biosimilar group *vs* 11 [6%] of 185 patients in the sorafenib group) and palmar-plantar erythrodysaesthesia syndrome (none *vs* 22 [12%]). 123 (32%) patients in the sintilimab–bevacizumab biosimilar group and 36 (19%) patients in the sorafenib group had serious adverse events. Treatment-related adverse events that led to death occurred in six (2%) patients in the sintilimab–bevacizumab biosimilar group (one patient with abnormal liver function, one patient with both hepatic failure and gastrointestinal haemorrhage, one patient with interstitial lung disease, one patient with both hepatic faliure and hyperkalemia, one patient with upper gastrointestinal haemorrhage, and one patient with intestinal volvulus) and two (1%) patients in the sorafenib group (one patient with gastrointestinal haemorrhage and one patient with death of unknown cause).

Interpretation Sintilimab plus IBI305 showed a significant overall survival and progression-free survival benefit versus sorafenib in the first-line setting for Chinese patients with unresectable, HBV-associated hepatocellular carcinoma, with an acceptable safety profile. This combination regimen could provide a novel treatment option for such patients.

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For the Chinese translation of the abstract see Online for appendix 1

*Study group members are listed in appendix 2 (p 2)

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Introduction

Hepatocellular carcinoma is the main type of primary liver cancer and the fourth leading cause of cancer-related death worldwide.[1] Infection with hepatitis B virus (HBV) is the predominant causative factor for hepatocellular carcinoma, and China has a high prevalence of HBV infection, with estimated new cases and deaths accounting for around half of global cases and deaths.[2]

Sorafenib and lenvatinib are multikinase inhibitors that are approved for first-line treatment of patients with advanced hepatocellular carcinoma. However, the efficacy of targeted agents is modest, and they confer limited survival benefits.[3–5] Moreover, the efficacy of sorafenib in patients with HBV-associated hepatocellular carcinoma was shown to be inferior to that in patients without HBV infection.[3,5]

Immunotherapies, including PD-1 and PD-L1 inhibitors, have shown clinical benefits in various cancers,[6,7] and several studies have indicated encouraging clinical activity of the anti-PD-1 antibody in previously treated patients with hepatocellular carcinoma.[8–10] However, several phase 3 studies did not show superiority of anti-PD-1 monotherapy compared with standard of care for the first-line or second-line treatment of hepatocellular carcinoma.[11,12]

In addition to having a role in angiogenesis, VEGF inhibitors have an attenuating effect on VEGF-mediated immunosuppression.[13] Combination treatment with an immune checkpoint inhibitor and an anti-VEGF antibody has been an effective strategy for the treatment of many solid tumours, including hepatocellular carcinoma.[14] The IMbrave 150 study showed a superior clinical benefit with atezolizumab plus bevacizumab over sorafenib monotherapy as first-line treatment for unresectable hepatocellular carcinoma.[15] Based on the results of that study, atezolizumab–bevacizumab was recently approved by the US Food and Drug Administration, the National Medical Products Administration of China (NMPA), the European Medicines Agency, and other regulatory agencies in the first-line setting for unresectable hepatocellular carcinoma. Considering differences in the treatment efficacy of PD-1 and PD-L1 inhibitors across various tumour types,[16] combination treatment with an anti-PD-1 antibody and an anti-angiogenesis agent might be a potential first-line treatment for patients with hepatocellular carcinoma.

Sintilimab is a selective anti-PD-1 antibody that inhibits the interaction between PD-1 and its ligands,[17] and has been approved by the NMPA for the treatment of relapsed or refractory classical Hodgkin lymphoma and for the treatment of unresectable locally advanced or metastatic non-squamous non-small-cell lung cancer when combined with pemetrexed and platinum.[18–20]

We aimed to evaluate the efficacy and safety of sintilimab plus IBI305, a bevacizumab biosimilar approved by the NMPA based on similarity in terms of pharmacokinetics

Research in context

Evidence before this study

We searched PubMed from inception to Dec 1, 2020, to identify clinical studies published in English that investigated first-line treatments for unresectable hepatocellular carcinoma, using the key terms "hepatocellular carcinoma" OR "hepatocellular cancer" AND "phase 3" OR "phase III" AND "first-line". Abstracts presented at relevant congresses were also searched using the same strategy, limiting the results to the past 2 years. The search retrieved a few key studies that reported on the therapeutic value of targeted treatment with sorafenib and lenvatinib, a study that showed an improvement in overall survival with donafenib over sorafenib, a study that did not meet the primary endpoint of overall survial non-inferiority for brivanib compared with sorafenib, a study of nivolumab (Checkmate 459) that showed no significant overall survival benefit over sorafenib, and a study that investigated combination treatment with a PD-L1 inhibitor and an anti-angiogenesis agent (IMbrave 150) that showed promising efficacy with improved overall survival and progression-free survival in patients with unresectable hepatocellular carcinoma. Based on this study, atezolizumab plus bevacizumab has been approved by the US Food and Drug Administration, European Medicines Agency, and the National Medical Products Administration of China as a first-line treatment for patients with systemic treatment-naive, unresectable or metastatic hepatocellular carcinoma.

Added value of this study

To our knowledge, ORIENT-32 is the first phase 3 study to show the superiority of an anti-PD-1 antibody (sintilimab) in combination with an anti-VEGF antibody (IBI305, a bevacizumab biosimilar) versus sorafenib in a first-line setting for patients with hepatocellular carcinoma. In contrast to the IMbrave 150 study, ORIENT-32 enrolled mostly patients with hepatitis B virus (HBV)-positive hepatocellular carcinoma and showed the survival benefit of a PD-1 inhibitor plus a VEGF inhibitor in such patients.

Implication of all the available evidence

The results of our study support the potential use of sintilimab in combination with a bevacizumab biosimilar as a first-line treatment option for Chinese patients with unresectable, HBV-associated hepatocellular carcinoma.

and efficacy to a reference drug (Avastin; Roche Genentech; San Francisco, CA, USA),[21] versus sorafenib as a first-line treatment for patients with unresectable hepatocellular carcinoma.

Methods

Study design and participants

ORIENT-32 was a randomised, open-label, phase 2–3 study at 50 clinical sites in China (appendix 2 pp 2–4). The phase 2 part of the study was a single-arm safety run-in, in which patients received only the sintilimab–bevacizumab biosimilar combination. The phase 3 part was a randomised, controlled trial in which patients were randomly assigned to either sintilimab–bevacizumab biosimilar or sorafenib.

The main inclusion criteria for both phase 2 and phase 3 were as follows: patients aged 18 years or older, with unresectable, locally advanced, or metastatic hepatocellular carcinoma, with a diagnosis that was histologically, cytologically, or clinically confirmed per the American Association for the Study of Liver Disease criteria,[22] who had received no previous systemic therapy for advanced or metastatic disease, had measurable disease per the Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1, had Barcelona Clinic Liver Cancer (BCLC) stage C or were unsuitable for radical surgery or had local treatment stage B, had a baseline Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1, had a Child-Pugh liver function score of 7 or less, and had adequate haematological and organ function with a life expectancy exceeding 12 weeks. Laboratory tests at baseline required to assess eligibility included absolute neutrophil count, platelet count, haemoglobin, total bilirubin, alanine aminotransferase, aspartate aminotransferase, serum creatinine, urine protein, and coagulation function.

Key exclusion criteria included a history of autoimmune disease, liver transplantation, acute or chronic active HBV or hepatitis C virus (HCV) infection, HBV DNA greater than 2000 IU/mL or 10^4 copies per mL, HCV RNA greater than 10^3 copies per mL, hepatitis B surface antigen and anti-HCV antibody positive concurrently (nucleotide antiviral treatment was permitted), CNS metastasis, involvement of both the main portal vein and the left and right branches by portal vein tumour thrombus, or of both the main trunk and the superior mesenteric vein concurrently, presence of tumour thrombus of inferior vena cava, known severe varicose veins assessed by endoscopy, or evidence of portal hypertension with a high risk of bleeding, previous treatments with any monoclonal antibody, radiotherapy (within 3 weeks), immunosuppressants or local therapies (within 4 weeks), or active traditional Chinese medicine (within 2 weeks) before the first dose of study treatment. Other systemic tumour therapies and locoregional treatments for target lesions were not allowed, but local palliative radiotherapy and bone metastasis treatment were permitted for isolated lesions (non-liver lesions) to control symptoms. Full eligibility criteria are provided in the study protocol (appendix 2).

The trial was done in accordance with Good Clinical Practice and the Declaration of Helsinki. The protocol and any amendments were approved by the institutional review board or ethics committee at each site. All patients provided written, informed consent before enrolment. The protocol is available in appendix 2.

Randomisation and masking

In the phase 3 part of the trial, eligible patients were randomly assigned (2:1) to receive either sintilimab plus IBI305 (sintilimb–bevacizumab biosimilar group) or sorafenib. Patients were stratified by macrovascular invasion or extrahepatic metastasis (yes *vs* no), baseline α-fetoprotein level (<400 ng/mL *vs* ≥400 ng/mL), and ECOG performance status (0 *vs* 1). Patients were randomly assigned using permuted block randomisation (block size of six) via an interactive web response system. The randomisation sequence was generated by an independent statistician, and the investigators obtained the randomisation assignments from the system after entering the stratification factors. The study was open-label and treatment allocation was not masked.

Procedures

In the phase 2 part of the trial (safety run-in period), patients received 200 mg of sintilimab intravenously over 60 min, followed by 15 mg/kg bodyweight of IBI305 bevacizumab biosimilar intravenously over 90 min (the second infusion over 60 min, and afterwards over 30 min if no infusion reaction occurred), every 3 weeks. The treatment continued until disease progression, development of intolerable toxic reactions, initiation of new anti-tumour treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring the termination of treatment stipulated in the protocol, whichever occurred first. Patients received sintilimab–bevacizumab biosimilar for a maximum of 24 months.

In the phase 3 part of the study, patients randomly allocated to the sintilimab–bevacizumab biosimilar group received 200 mg of sintilimab intravenously over 60 min, followed by 15 mg/kg bodyweight of IBI305 bevacizumab biosimilar intravenously over 90 min (the second infusion over 60 min, and afterwards over 30 min if no infusion reaction occurred), every 3 weeks. Patients randomly allocated to the sorafenib group received 400 mg of sorafenib orally twice daily. Patients received their assigned treatment until disease progression, intolerable toxicity, or withdrawal of consent. Patients in both groups could continue treatment beyond disease progression at the investigator's discretion for clinical benefit. Dose modifications were permitted in the sorafenib group (from 400 mg twice daily to 400 mg once daily, and then could be reduced to 400 mg every other day), but were not

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See **Online** for appendix 2

allowed in the sintilimab–bevacizumab biosimilar group. The dose of bevacizumab biosimilar was calculated according to the patient's bodyweight at baseline (before the first dose) and remained unchanged during the study, unless the patient's weight varied by 10% or greater from baseline. In the sintilimab–bevacizumab biosimilar group, patients with interruption or discontinuation of either sintilimab or bevacizumab biosimilar due to adverse events were permitted to continue bevacizumab biosimilar or sintilimab monotherapy. Patients received sintilimab–bevacizumab biosimilar for a maximum of 24 months.

Tumour assessments were done in hospital by contrast CT or MRI at baseline and every 6 weeks until week 48, and then every 12 weeks. In the phase 2 part of the trial, tumour response was assessed by the investigator per RECIST 1.1 only. In the phase 3 parts of the study, tumour response was assessed by an independent radiological review committee (IRCC), who were masked to assigned treatment and outcomes. Both RECIST 1.1 and hepatocellular carcinoma-specific modified RECIST (mRECIST) criteria were used in the phase 2 and phase 3 part of the study. RECIST 1.1 was used by the investigator and IRRC, but mRECIST was used only by the IRRC. Because of the possibility of pseudoprogression in patients treated with sintilimab, the confirmation of progression based on immune-related RECIST after 4–6 weeks was required for patients who had reported disease progression for the first time. Safety was evaluated from consent through the follow-up period. The incidence and severity of adverse events were assessed by investigators according to the National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events (CTCAE) version 5.0. Adverse events, including treatment-related adverse events or serious adverse events, were assessed throughout the treatment period and for up to 90 days after the last dose. The main types of laboratory tests were monitored at baseline, before study drug administration on a 3-week schedule, and during the safety follow-up period. A safety follow-up was done within 30 days (± 3 days) after the last dose or before the start of a new antitumour treatment, whichever occurred first. All adverse events, including serious adverse events, that occurred from the signing of the informed consent form to 30 days after the last dose were collected, regardless of whether they were observed by the investigator or self-reported by the patient. From day 30 to day 90 after the last dose, the investigator reported all serious adverse events and adverse events related to study drug or procedures. A survival follow-up was done every 60 days (± 7 days) after the safety follow-up.

Patient-reported outcomes were assessed by the European Organisation for Research and Treatment of Cancer (EORTC) Quality-of-life Questionnaire Core 30 (QLQ-C30) and Quality-of-life Questionnaire-Hepatocellular Carcinoma 18 (QLQ-HCC18) at the same time points as the tumour imaging assessments.

Immunogenicity of sintilimab in the sintilimab–bevacizumab biosimilar group was also assessed using blood samples taken at baseline and after treatment (appendix 2 p 1). Immunogenicity samples were collected within 1 h before sintilimab infusion in cycles one, two, four, eight, twelve, and sixteen, and then every eight cycles (cycle 24, 32, and so on) thereafter, and during safety follow-up.

Outcomes

The primary endpoint of the phase 2 part of the study was the safety and tolerability of combination treatment. Secondary endpoints were objective response rate (the percentage of patients whose best overall response was complete response or partial response), disease control rate (the proportion of patients who had a complete response, partial response, or stable disease), duration of response (the time from the first documented complete response or partial response to disease progression or death), time to progression (the time from the first dose of study drug to the first documented progression), time to response (the time from the first dose of study drug to the first confirmed complete response or partial response), and progression-free survival (the time from the first dose of study drug to the first documented disease progression or death from any cause) assessed by the investigator per RECIST version 1.1, overall survival (the time from the first dose of study drug to death from any cause), pharmacokinetics, and immunogenicity of sintilimab–bevacizumab biosimilar.

For the phase 3 part of the study, the co-primary endpoints were overall survival and progression-free survival as assessed by the IRRC per RECIST version 1.1. The secondary endpoints were investigator-assessed progression-free surival per RECIST version 1.1 and objective response rate, disease control rate, time to response, time to progression, and duration of response assessed by both IRRC and investigators per RECIST version 1.1. Progression-free survival, objective response rate, disease control rate, duration of response, time to response, and time to progression were also assessed by the IRRC according to hepatocellular carcinoma-specific modified RECIST (mRECIST). Secondary endpoints also included time to deterioration of global health status from QLQ-C30, defined as a decrease of 10 points or more from baseline, and the immunogenicity of sintilimab in the sintilimab–bevacizumab biosimilar group. A post-hoc analysis was done for time to deterioration on eight domains from QLQ-HCC18, with a similar definition to time to deterioration of global health status. The eight domains are pyrexia, sex life, fatigue, pain, abdominal distension, nutrition, physical function, and jaundice.

Statistical analysis

20 patients were planned to be enrolled in the phase 2 part of the study. It was predicted that the safety profile of sintilimab in combination with IBI305 would be similar

to that of atezolizumab plus bevacizumab (ie, treatment-related adverse events at grade 3 or worse as per CTCAE were observed in 17 [25%; 95% CI 15–37] of 68 patients receiving atezolizumab plus bevacizumab).[22] Assuming that more than eight (40%) of 20 patients receiving sintilimab–bevacizumab biosimilar have treatment-related adverse events at grade 3 or worse as per CTCAE, there is about 90% power in inferring that the true incidence of such adverse events exceeds 29%. Since the actual number of patients screened was more than the planned enrollment number, a total of 24 patients were enrolled in the phase 2 study. Analyses in the phase 2 part of the study were descriptive for all endpoints and done in the safety set, defined as patients who have received at least one dose of study drug.

Following discussion by the safety assessment committee, based on the preliminary safety and activity data of the phase 2 part of the study (grade 3 or worse treatment-related adverse events in 29% of patients), the phase 3, randomised, controlled part of study was carried out.

For the phase 3 part of the study, we planned to enrol 546 patients in whom we expected there would be 408 deaths, providing a power of 89% to detect a hazard ratio (HR) of 0·714 favouring sintilimab–bevacizumab biosimilar over sorafenib, with a two-sided α=0·048.

A first interim analysis of overall survival was planned when 265 (65%) deaths were observed. The cutoff value of the interim analysis approximates the O'Brien-Fleming boundary according to the Lan-Demets method. The interim analysis has a two-sided level of significance α=0·0102. The primary analysis of progression-free survival was planned to be done concurrently with the first interim analysis of overall survival, when approximately 400 progression-free survival events had occurred. The two-sided significance level for progression-free survival was 0·002 with more than 99% power. The family-wise type I error rate for this study was strictly controlled at 0·05 (two-sided). The Bonferroni correction method was used to adjust α to control the overall type I error rate. The progression-free survival endpoint was analysed first; if the test was significant, the two-sided level of significance α=0·002 allocated to progression-free survival would be recovered to the overall survival hypothesis test (ie, α=0·05). If the test results of progression-free survival were not significant, overall survival would be tested at the original significance level α=0·048.

If the number of deaths was fewer than 220 when 400 progression-free survival events had occurred, two interim analyses were planned for overall survival: the first after 400 progression-free survival events and the second when 265 deaths occurred.

Efficacy analyses in the phase 3 part of the trial for overall survival and progression-free survival included all patients randomly assigned to a group according to the intention-to-treat principle, regardless of whether or not they received treatment. Efficacy analyses for objective response rate were done in both the full analysis set and response-evaluable population who had at least one tumour assessment or died before the first scheduled tumour assessment. Because of COVID-19 quarantine, tumour assessments were not done for some patients. Therefore, the response-evaluable population might better reflect actual efficacy. Objective response rate results in the full analysis set are in appendix 2 (p28). Patients who died before the first tumour assessment (ie, those who died within 6 weeks after randomisation) were considered to have poor treatment efficacy, and thus were included in the analysis of objective response rate among the response-evaluable population as non-responders. Safety analyses included all patients who received at least one dose of study treatment.

We used a stratified log-rank test to compare the time-to-event endpoints (overall survival and progression-free survival) between treatment groups. We calculated HRs and associated 95% CIs based on a stratified Cox proportional hazards model. Stratification factors were the same as those used for randomisation. We used the Kaplan-Meier method to estimate the median overall survival, progression-free surival, and 95% CIs and plot survival curves. To assess the association between co-primary endpoints and stratification factors or baseline variables, prespecified subgroup analyses were done based on the factors macrovascular invasion or extrahepatic metastasis (yes *vs* no), α-fetoprotein level (<400 ng/mL *vs* ≥400 ng/mL), ECOG performance status (0 *vs* 1), macrovascular invasion (yes *vs* no), extrahepatic metastasis (yes *vs* no), BCLC stage (B *vs* C), Child-Pugh class (A *vs* B), HBV status (positive *vs* negative), previous transarterial chemoembolisation (yes *vs* no), previous local ablation (yes *vs* no), and EORTC QLQ-C30 score (global health status scale at baseline; less than or equal to median *vs* greater than median). Given the exploratory nature of the subgroup analyses and potential small sample sizes, the HRs and corresponding 95% CIs were estimated from an unstratified Cox model. We checked the proportional hazards assumption in the Cox model using three methods (visual inspection, including time dependent covariates in the Cox model, and Schoenfeld residuals) on the co-primary endpoints (appendix 2 p 1).

The Clopper-Pearson method was used to calculate the 95% CIs for the objective response rate and disesase control rate for each treatment group. We used the Cochran-Mantel-Haenszel test and stratified Miettinen and Nurminen method to compare the objective response rate and disease control rate between treatment groups. The stratification factors were the same as those used to analyse overall survival and progression-free surival. Duration of response, time to response, time to progression, and time to deterioration of quality of life were estimated by the Kaplan-Meier method. A post-hoc analysis of the duration of disease control was done in both

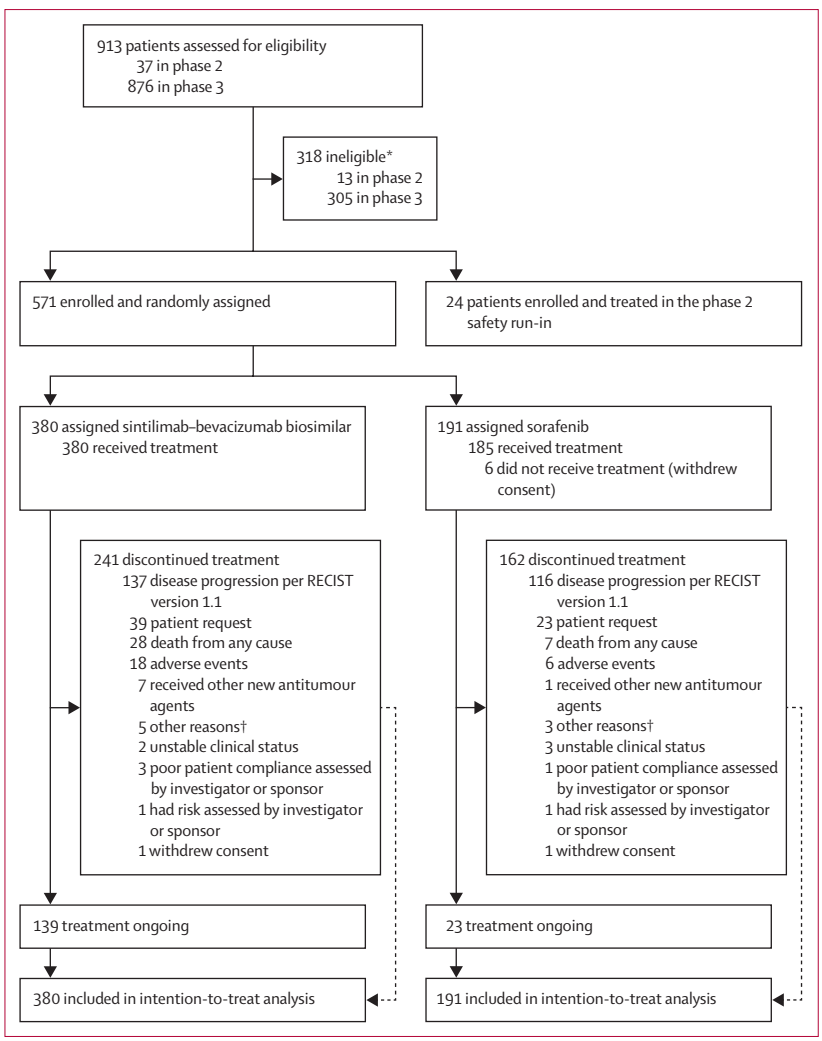


Figure 1: **Trial profile**
RECIST=Response Evaluation Criteria in Solid Tumors. *Details are shown in the appendix (pp 5–7). †Other reasons included protocol deviation or non-compliance with the protocol.

	Sintilimab–bevacizumab biosimilar group (n=380)	Sorafenib group (n=191)
Age, years	53 (21–82)	54 (28–77)
Sex		
Male	334 (88%)	171 (90%)
Female	46 (12%)	20 (10%)
Eastern Cooperative Oncology Group performance status		
0	183 (48%)	91 (48%)
1	197 (52%)	100 (52%)
Child-Pugh classification		
A (5–6)	365 (96%)	182 (95%)
B (7)	15 (4%)	9 (5%)
Barcelona Clinic Liver Cancer stage		
B	56 (15%)	27 (14%)
C	324 (85%)	164 (86%)
α-fetoprotein concentration		
<400 ng/mL	215 (57%)	110 (58%)
≥400 ng/mL	165 (43%)	81 (42%)
Presence of macrovascular invasion, extrahepatic metastasis, or both	303 (80%)	150 (79%)
Macrovascular invasion	105 (28%)	50 (26%)
Extrahepatic metastasis	279 (73%)	144 (75%)
Hepatitis B virus infection	359 (94%)	179 (94%)
Hepatitis C virus infection	6 (2%)	8 (4%)
Previous local therapy for hepatocellular carcinoma at least once	308 (81%)	162 (85%)
Ablation	95 (25%)	45 (24%)
Surgery	199 (52%)	101 (53%)
Radiotherapy	12 (3%)	15 (8%)
Transarterial chemoembolisation	250 (66%)	127 (66%)

Data are median (range) or n (%).

Table 1: **Baseline characteristics of patients in the intention-to-treat population (phase 3)**

of the study groups by investigators using the Kaplan-Meier method to further analyse treatment benefit.

All statistical analyses were done using SAS version 9.4. An independent data monitoring committee reviewed safety data. The study is registered with ClinicalTrials.gov, NCT03794440.

Role of the funding source

The funder of the study worked with the investigators to design the study and to collect, analyse, and interpret the data. The funder and JF and ZR were responsible for study oversight. ZR, JF, YY, MC, YW, and HZ prepared the manuscript draft, and all co-authors participated in data collection and reviewing the manuscript.

Results

Between Feb 11, 2019, and Jan 15, 2020, we screened 913 patients (37 for phase 2 and 876 for phase 3). After exclusion of 318 patients who did not meet the eligibility criteria (appendix 2 pp 5–7), 595 patients were enrolled (24 in phase 2 and 571 in phase 3). In the phase 2 part of the study, 37 patients were screened, of whom 24 were enrolled and treated.

As of Aug 15, 2020, the median overall survival follow-up duration reached 15·8 months (IQR 15·2–16·1). In phase 2, treatment-emergent adverse events related to either study drug were reported in 22 (92%) of 24 patients, with an incidence of grade 3–4 treatment-related adverse events of 29% (seven of 24 patients). Treatment-emergent serious adverse events occurred in seven (29%) patients, including six (25%) patients with treatment-related serious adverse events; no adverse events leading to death occurred. The objective response rate assessed by the investigator per RECIST 1.1 was 25·0%. The data from phase 2 part showed an acceptable safety profile. Complete phase 2 results are shown in appendix 2 (pp 31–37).

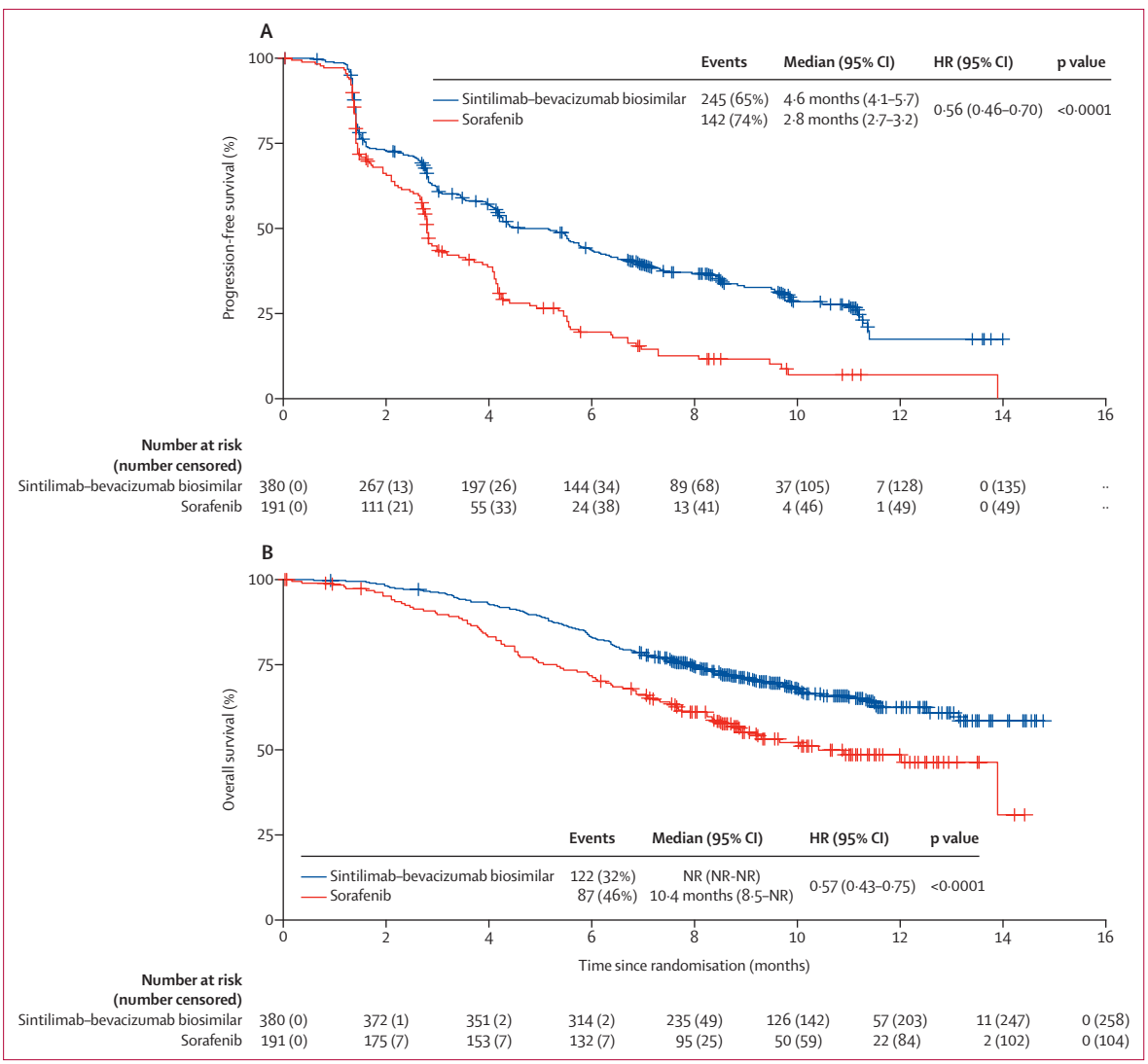


Figure 2: **Kaplan–Meier analysis of IRRC-assessed progression-free survival according to RECIST version 1.1 (A) and overall survival (B) in the intention-to-treat population**

Vertical bars indicate censored events (death or progression). HR=hazard ratio. IRRC=independent radiological review committee. NR=not reached. RECIST=Response Evaluation Criteria in Solid Tumors.

In the phase 3 part of the study, 571 eligible patients were randomly assigned to either the sintilimab–bevacizumab biosimilar group (380 patients) or sorafenib group (191 patients). All patients in the sintilimab–bevacizumab biosimilar group and 185 patients in the sorafenib group received at least one dose of study treatment (figure 1). Demographic and disease characteristics at baseline were similar between the two study groups (table 1).

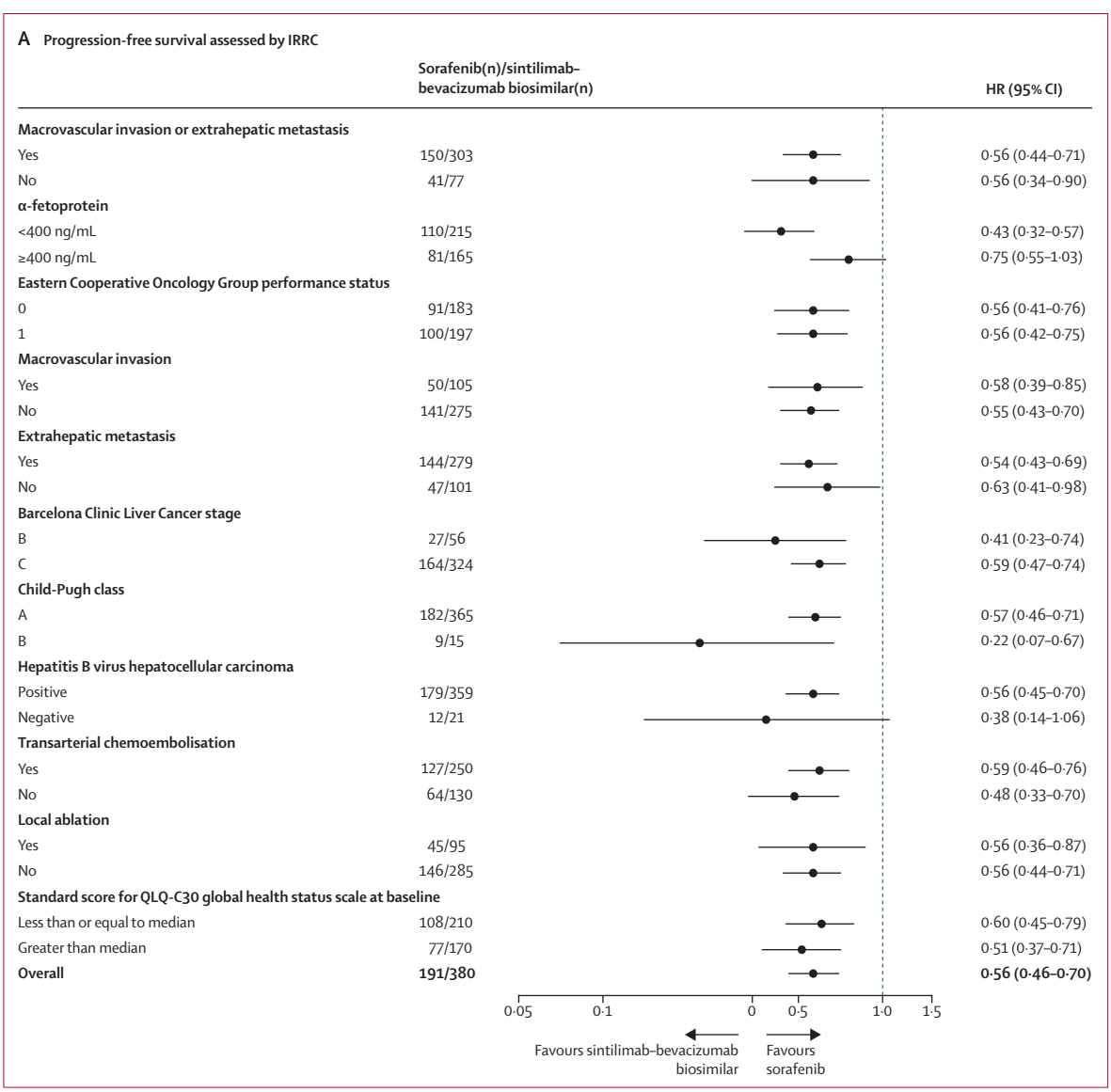
At data cutoff (August 15, 2020), median follow-up was 10·0 months (IQR 8·5–11·7) in the sintilimab–bevacizumab biosimilar group and 10·0 months (8·4–11·7) in the sorafenib group. 245 (64%) of 380 patients in the sintilimab–bevacizumab biosimilar group and 142 (74%) of 191 patients in the sorafenib group had progressive disease, as assessed by the IRRC per RECIST version 1.1, or had died. Patients in the sintilimab–bevacizumab biosimilar group had a significantly longer IRRC-assessed median progression-free survival (4·6 months [95% CI 4·1–5·7]) than did patients in the sorafenib group (2·8 months [2·7–3·2]; stratified HR 0·56, 95% CI 0·46–0·70; p<0·0001; figure 2A). In the first interim analysis of overall survival, 122 (32%) of 380 patients in the sintilimab–bevacizumab biosimilar group and 87 (46%) of 191 patients in the sorafenib group had died. Sintilimab–bevacizumab biosimilar significantly prolonged overall survival compared with sorafenib (stratified HR 0·57, 95% CI 0·43–0·75; p<0·0001). Median overall survival was not reached (95% CI not reached–not reached) in the sintilimab-bevacizumab biosimilar group versus

10·4 months (8·5–not reached) in the sorafenib group (HR 0·57, 95% CI 0·43–0·75; p<0·0001; figure 2B).

Predefined subgroup analyses of progression-free survival assessed by IRRC and overall survival are shown in figure 3. Clinical benefits were consistent across most predefined subgroups both for progression-free and overall survival.

In the response-evaluable population, the proportion of patients with a confirmed objective response was significantly higher in the sintilimab–bevacizumab group than in the sorafenib group by IRRC assessment (RECIST version 1.1 and mRECIST) and by investigator assessment (RECIST version 1.1; table 2). Confirmed disease control and duration of response by IRRC and investigator (per RECIST 1.1) and by IRRC per mRECIST are also shown in table 2. The post-hoc analysis of the duration of disease control is shown in table 2 and appendix 2 (p 16). Results for median time to response and median time to progression assessed by the IRRC and investigator according to RECIST version 1.1 and by the IRRC according to mRECIST are shown in appendix 2 (pp 7–8). Progression-free survival by investigator (RECIST 1.1) and IRRC (mRECIST) are also shown in appendix 2 (pp 7–8).

Patients in the sintilimab–bevacizumab biosimilar group had a significantly longer time to deterioration of global health status (6·7 months, 95% CI 5·5–7·3) than did patients in the sorafenib group (4·1 months, 2·9–5·2; stratified HR 0·73, 95% CI 0·56–0·94 (appendix 2 p 17). Time to deterioration of the QLQ-HCC18 domains (except jaundice) also showed delayed deterioration in the sintilimab–bevacizumab biosimilar group compared with the sorafenib group (appendix 2 pp 18–25).



(Figure 3 continues on next page)

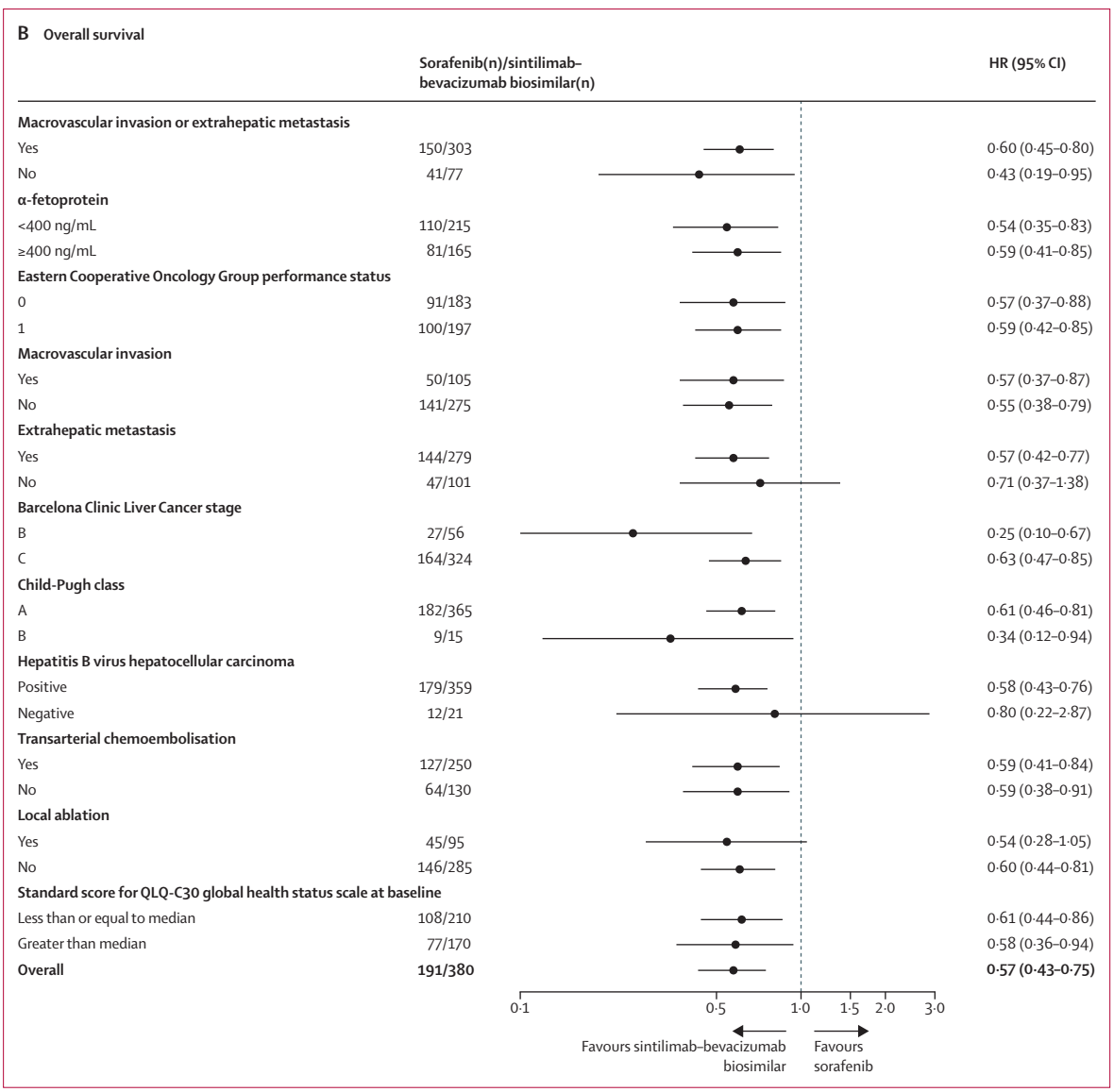


Figure 3: **IRRC-assessed progression-free survival (A) and overall survival (B) in prespecified subgroups**

109 (29%) of 380 patients in the sintilimab–bevacizumab biosimilar group and 89 (47%) of 191 patients in the sorafenib group received subsequent therapies (appendix 2 p 29), including nine (2%) patients in the sintilimab–bevacizumab biosimilar group and 40 (21%) patients in the sorafenib group who received anti-PD-1 or anti-PD-L1 agents.

565 patients received at least one dose of study treatment and were included in the safety analyses (380 patients in the sintilimab–bevacizumab biosimilar group and 185 patients in the sorafenib group). Patients in the sintilimab–bevacizumab biosimilar group had a median duration of treatment of 7·0 months (range 0·7–15·2) for sintiliamb and 6·6 months (range 0·7–15·2) for bevacizumab biosimilar, and patients in the sorafenib group had a median duration of treatment of 3·5 months (range 0·02–14·2) for sorafenib. The median relative dose intensities of sintilimab, bevacizumab biosimilar, and sorafenib were 93% (range 33–108), 94% (range 32–108), and 93% (range 26–100), respectively.

Study treatment was interrupted due to treatment-emergent adverse events in 188 (49%) of 380 patients in the sintilimab–bevacizumab biosimilar group and 75 (41%) of 185 patients in the sorafenib group (appendix 2 p 8). 66 (36%) patients in the sorafenib group had a dose reduction because of a treatment-emergent adverse event. Treatment was discontinued due to a treatment-emergent adverse event in 52 (14%) patients in the sintilimab–bevacizumab biosimilar group and 11 (6%) patients in the sorafenib group. Bleeding events leading

	IRRC-assessed RECIST version 1.1			Investigator-assessed RECIST version 1.1			IRRC-assessed modified RECIST		
	Sintilimab-bevacizumab biosimilar group (n=365)	Sorafenib group (n=172)	p value*	Sintilimab-bevacizumab biosimilar group (n=367)	Sorafenib group (n=173)	p value*	Sintilimab-bevacizumab biosimilar group (n=366)	Sorafenib group (n=172)	p value*
Confirmed objective response†	75‡ (21%; 17–25)	7 (4%; 2–8)	<0·0001	72 (20%; 16–24)	5 (3%; 1–7)	<0·0001	89§ (24%; 20–29)	13 (8%; 4–13)	<0·0001
Complete response	0	0	··	2 (1%)	0	··	2 (1%)	0	··
Partial response	75 (21%)	7 (4%)	··	70 (19%)	5 (3%)	··	87 (24%)	13 (8%)	··
Stable disease	189 (52%)	103 (60%)	··	195 (53%)	111 (64%)	··	179 (49%)	96 (56%)	··
Progressive disease	98 (27%)	57 (33%)	··	98 (27%)	51 (29%)	··	95 (26%)	58 (34%)	··
Not evaluable¶	1 (<1%)	1 (<1%)	··	0	2 (1%)	··	1 (<1%)	1 (1%)	··
Not done‖	2 (1%)	4 (2%)	··	2 (1%)	4 (2%)	··	2 (1%)	4 (2%)	··
Confirmed disease control**	264 (72%; 67–77)	110 (64%; 56–71)	··	267 (73%; 68–77)	116 (67%; 60–74)	··	268 (73%; 68–78)	109 (63%; 56–71)	··
Duration of response, months	NE (NE–NE)	9·8 (2·8–NE)	··	NE (NE–NE)	NE (2·5–NE)	··	NE (8·2–NE)	6·6 (2·6–NE)	··
Duration of disease control, months	7·0 (5·1–8·2)	2·9 (2·6–3·7)	··	6·9 (5·6–8·4)	3·5 (2·8–4·4)	··	5·8 (4·8–7·2)	2·9 (2·4–3·7)	··

Data are n (%; 95% CI), n (%), or median (95% CI), unless stated otherwise. 95% CI was calculated using the Clopper-Pearson method. Response-evaluable patients were those who had measurable disease according to RECIST version 1.1 assessed by the IRRC or investigator, and according to hepatocellular carcinoma-specific modified RECIST by the IRRC. IRRC=independent radiological review committee. NE=not evaluable. RECIST=Response Evaluation Criteria in Solid Tumors. *p value was calculated using the stratified Miettinen and Nurminen method. †Confirmed objective response was defined as a response (complete or partial) confirmed by two consecutive tumour assessments with at least a 28-day interval. ‡Three patients with two consecutive partial responses across the cutoff date were included. §Two patients with two consecutive partial responses after the cutoff date were included. ¶Inadequate data for evaluation. ‖The imaging assessment was not done at the first tumour assessment (week 6). **Disease control included complete response, partial response, and stable disease.

Table 2: **Response and duration of response in response-evaluable patients**

to treatment discontinuation were upper gastrointestinal haemorrhage (seven [2%] in the sintilimab–bevacizumab biosimilar group *vs* none in the sorafenib group), oesophageal variceal bleeding (two [<1%] *vs* none), and gastrointestinal haemorrhage (one [<1%] *vs* two [<1%]; appendix 2 pp 8–12).

376 (99%) of 380 patients in the sintilimab–bevacizumab biosimilar group and 181 (98%) of 185 patients in the sorafenib group had treatment-emergent adverse events of any grade (table 3; appendix 2 p 8). The most frequent grade 3 or worse treatment-emergent adverse events (occuring in ≥5% of patients) were hypertension (55 [15%] of 380 patients in the sintilimab–bevacizumab biosimilar group *vs* 11 [6%] of 185 patients in the sorafenib group), decreased platelet count (31 [8%] patients *vs* five [3%] patients), proteinuria (20 [5%] patients *vs* three [2%] patients), increased γ-glutamyltransferase (19 [5%] patients *vs* three [2%] patients, increased aspartate aminotransferase (seven [2%] *vs* 10 [5%] patients), and palmar-plantar erythrodysaesthesia syndrome (no patients *vs* 22 [12%] patients; table 3). Treatment-related adverse events are presented in appendix 2 (pp 12–13). Serious adverse events were reported in 123 (32%) of 380 patients in the sintilimab–bevacizumab biosimilar group versus 36 (19%) of 185 patients in the sorafenib group (appendix 2 p 14). Treatment-emergent adverse events leading to death occurred in ten (3%) patients in the sintilimab–bevacizumab biosimilar group and in six (3%) patients in the sorafenib group. Treatment-related adverse events occurred in six (2%) patients in the sintilimab–bevacizumab biosimilar group (one patient with abnormal liver function, one patient with both hepatic failure and gastrointestinal haemorrhage, one patient with interstitial lung disease, one patient with both hepatic failure and hyperkalemia, one patient with upper gastrointestinal haemorrhage, and one patient with intestinal volvulus) and two (1%) patients in the sorafenib group (one patient with gastrointestinal haemorrhage and one patient with death of unknown cause). Adverse events of special interest in either group are presented in appendix 2 (pp 14–15).

In the sintilimab–bevacizumab biosimilar group, among 363 patients who had evaluable serum samples, 12 (3%) tested positive for treatment-emergent antidrug antibodies against sintilimab (appendix 2 p 15).

This study was affected by COVID-19. 139 (37%) of 380 patients in the sintilimab–bevacizumab biosimilar group and 36 (19%) of 185 patients in the sorafenib group were affected by COVID-19: drug administration was delayed by more than one cycle (>21 days) for 99 (26%) patients in the sintilimab–bevacizumab group and by more than 7 days for 16 (9%) patients in the sorafenib group; radiographic assessment of tumours was missed or delayed by more than 6 weeks in 89 (23%) patients in the sintilimab–bevacizumab biosimilar group and in 28 (15%) patients in the sorafenib group. To our knowledge, no patients were infected with SARS-CoV-2.

Discussion

In this study, we observed significant improvements in overall survival and progression-free surival after combination treatment with sintilimab–bevacizumab biosimilar

	Sintilimab–bevacizumab biosimilar group (n=380)				Sorafenib group (n=185)			
	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5
Any treatment-emergent adverse event	167 (44%)	181 (48%)	18 (5%)	10 (3%)	92 (50%)	79 (43%)	4 (2%)	6 (3%)
Proteinuria	140 (37%)	20 (5%)	0	0	31 (17%)	3 (2%)	0	0
Increased aspartate aminotransferase	128 (34%)	6 (2%)	1 (<1%)	0	65 (35%)	10 (5%)	0	0
Decreased platelet count	124 (33%)	26 (7%)	5 (1%)	0	50 (27%)	4 (2%)	1 (1%)	0
Increased alanine aminotransferase	94 (25%)	4 (1%)	1 (<1%)	0	47 (25%)	5 (3%)	0	0
Increased blood bilirubin	93 (24%)	17 (4%)	2 (1%)	0	60 (32%)	6 (3%)	0	0
Decreased white blood cell count	71 (19%)	6 (2%)	0	0	26 (14%)	3 (2%)	1 (1%)	0
Hypertension	66 (17%)	55 (14%)	0	0	19 (10%)	11 (6%)	0	0
Pyrexia	65 (17%)	1 (<1%)	0	0	19 (10%)	1 (1%)	0	0
Hypoalbuminaemia	63 (17%)	0	0	0	17 (9%)	1 (1%)	0	0
Asthenia	58 (15%)	3 (1%)	0	0	34 (18%)	2 (1%)	0	0
Decreased weight	55 (14%)	2 (1%)	0	0	31 (17%)	0	0	0
Abdominal pain	53 (14%)	4 (1%)	0	0	25 (14%)	0	0	0
Hypothyroidism	53 (14%)	0	0	0	14 (8%)	0	0	0
Diarrhoea	49 (13%)	6 (2%)	0	0	72 (39%)	5 (3%)	0	0
Increased γ-glutamyltransferase	44 (12%)	19 (5%)	0	0	26 (14%)	3 (2%)	0	0
Increased blood thyroid-stimulating hormone	43 (11%)	0	0	0	17 (9%)	0	0	0
Abdominal distension	42 (11%)	2 (1%)	0	0	25 (14%)	1 (1%)	0	0
Increased blood alkaline phosphatase	42 (11%)	5 (1%)	0	0	13 (7%)	1 (1%)	0	0
Anaemia	41 (11%)	10 (3%)	0	0	13 (7%)	2 (1%)	0	0
Decreased appetite	41 (11%)	2 (1%)	0	0	22 (12%)	0	0	0
Pruritus	39 (10%)	0	0	0	10 (5%)	0	0	0
Nausea	26 (7%)	2 (1%)	0	0	19 (10%)	0	0	0
Rash	25 (7%)	1 (<1%)	0	0	21 (11%)	2 (1%)	0	0
Vomiting	25 (7%)	1 (<1%)	0	0	19 (10%)	0	0	0
Increased blood lactate dehydrogenase	20 (5%)	0	0	0	25 (14%)	0	0	0
Palmar-plantar erythrodysaesthesia syndrome	1 (<1%)	0	0	0	80 (43%)	22 (12%)	0	0
Alopecia	0	0	0	0	38 (21%)	0	0	0
Detection of proteinuria	34 (9%)	2 (1%)	0	0	15 (8%)	2 (1%)	0	0
Decreased neutrophil count	31 (8%)	8 (2%)	0	0	15 (8%)	2 (1%)	1 (1%)	0
Increased conjugated bilirubin	30 (8%)	10 (3%)	0	0	18 (10%)	2 (1%)	0	0
Elevated blood pressure	25 (7%)	10 (3%)	0	0	6 (3%)	3 (2%)	0	0
Decreased lymphocyte count	19 (5%)	7 (2%)	0	0	12 (6%)	2 (1%)	0	0
Ascites	8 (2%)	7 (2%)	0	0	4 (2%)	1 (1%)	0	0
Upper gastrointestinal haemorrhage	3 (1%)	7 (2%)	1 (<1%)	1 (<1%)	2 (1%)	1 (1%)	1 (1%)	0
Gastrointestinal haemorrhage	2 (1%)	3 (1%)	0	1 (<1%)	2 (1%)	2 (1%)	0	1 (1%)
Unknown death	0	0	0	2 (1%)	0	0	0	2 (1%)
Hyponatremia	24 (6%)	7 (2%)	0	0	7 (4%)	2 (1%)	1 (1%)	0
Hypokalemia	21 (6%)	5 (1%)	2 (1%)	0	13 (7%)	2 (1%)	0	0
Hyperglycemia	8 (2%)	4 (1%)	0	0	1 (1%)	0	0	0
Upper respiratory tract infection	22 (6%)	5 (1%)	0	0	7 (4%)	3 (2%)	0	0
Infectious pneumonia	10 (3%)	5 (1%)	0	0	3 (2%)	0	0	0
Hepatic abscess	0	0	0	0	0	2 (1%)	0	0
Backache	18 (5%)	4 (1%)	0	0	6 (3%)	1 (1%)	0	0
Abnormal liver function	11 (3%)	8 (2%)	3 (1%)	2 (1%)	5 (3%)	4 (2%)	1 (1%)	0
Jaundice	2 (1%)	1 (<1%)	0	0	0	2 (1%)	0	0
Hepatic failure	1 (<1%)	1 (<1%)	2 (1%)	2 (1%)	1 (1%)	0	0	2 (1%)

Data are n (%). Grade 1–2 treatment-emergent adverse events occurring in ≥10% of patients in either group and grade 3, 4, 5 treatment-emergent adverse events occurring in ≥1% of patients in either group are shown.

Table 3: **Treatment-emergent adverse events**

versus sorafenib in patients with unresectable, systemic treatment-naive hepatocellular carcinoma. Our results were consistent across all predefined subgroups and supported the results of a previous phase 1b study.[19] To our knowledge, ORIENT-32 is the first large-scale, phase 2–3 study of a PD-1 inhibitor in combination with an anti-angiogenesis inhibitor, and the findings contribute to understanding immunotherapies in combination with anti-VEGF therapies in patients with hepatocellular carcinoma. Compared with similar studies of first-line therapies for patients with advanced hepatocellular carcinoma,[3,4,15] sintilimab–bevacizumab biosimilar reduced the risk of death and disease progression. To our knowledge, this was the first large-scale study to show the benefit of first-line treatment in HBV-associated hepatocellular carcinoma, providing support for a clinical benefit with combined immunotherapy versus targeted therapy in this population.

The baseline characteristics of patients in the ORIENT-32 study differed from those in the IMbrave 150 study.[15] Patients in ORIENT-32 were younger (median age for patients treated with combination treatment was 53 years in ORIENT-32 *vs* 64 years in IMbrave 150; for those treated with sorafenib it was 54 years *vs* 66 years), had more patients with an ECOG performance status of 1 (52% *vs* 38%), had a higher incidence of extrahepatic metastasis (74% *vs* 61%), had a higher proportion of patients with high α-fetoprotein (≥400 ng/mL; 43% *vs* 38%), were much more heavily pretreated with local therapy (82% *vs* 49%), and included patients with Child–Pugh B, who were excluded from most other studies evaluating first-line therapies in patients with hepatocellular carcinoma.[4,11,15] Additionally, most patients in our study had HBV-associated hepatocellular carcinoma, compared with less than 50% of participants in the IMbrave 150 study.[15] These data suggest that patients with HBV-associated hepatocellular carcinoma in China are characterised by younger age, poor ECOG performance status, increased possibility of extrahepatic extension, high tumour burden, an increased likelihood of having received previous locoregional therapy, and poor liver function with cirrhosis compared with patients in high-income countries. Impaired liver function due to cirrhosis attenuates the treatment efficacy of the targeted therapy of sorafenib in patients with hepatocellular carcinoma in Asia.[3,5] For the past decade, there has been a high unmet treatment need in such patients.

In ORIENT-32, the IRRC-assessed objective response rate according to RECIST version 1.1 was significantly higher in patients in the sintilimab–bevacizumab biosimilar group than in patients in the sorafenib group (21% *vs* 4%), with lower values overall than those reported in the IMbrave 150 study (27% *vs* 12%) and its updated results (30% *vs* 11%).[13,23] The relatively low objective response rate and absence of complete reponse as assessed by the IRRC might be due to differences in the study populations. Chinese patients have a high prevalence of HBV-associated hepatocellular carcinoma, which might have a poorer prognosis compared with other hepatocellular carcinoma types. Additionally, the study was done during the COVID-19 pandemic, which led to delayed follow-up visits, which could explain the low objective response rate. The median duration of response was not reached in the sintilimab–bevacizumab biosimilar group. We analysed (post-hoc) duration of disease control, which included duration of benefit not only for responders, but also for patients who had stable disease; the prolonged duration of disease control in the combination treatment setting might contribute to the significant overall survival benefit we observed. The best way to measure the potential benefit of immunotherapy is not known, thus duration of disease control as a measure of benefit warrants further investigation.

Poor liver function is generally considered to be a risk factor for poor outcomes after tumour therapy, and is associated with a high risk of adverse events. In the Checkmate 040 study,[24] nivolumab showed an encouraging disease control rate (55%) and durable response (median duration of response 9·9 months) in patients with advanced hepatocellular carcinoma and Child-Pugh B status, compared with historical data. Similar to this finding, in the subgroup of patients with Child-Pugh class B in ORIENT-32, sintilimab–bevacizumab biosimilar treatment markedly reduced the risk of death and disease progression. However, patients with Child-Pugh class B only accounted for 4% of the population in the ORIENT-32 study, and further studies in patients with hepatocellular carcinoma and Child-Pugh class B are warranted.

The safety profile of the sintilimab–bevacizumab biosimilar combination was generally consistent with historical data on sintilimab[17,18] and bevacizumab,[21] with no new safety signals reported. 52 (14%) of 380 patients dicontinued sintilimab–bevacizumab biosimilar because of treatment-emergent adverse events, which was similar to the incidence previously reported with atezolizumab–bevacizumab treatment (16%).[15] Gastrointestinal events were the most frequent reasons for discontinuation of the bevacizumab biosimilar. In IMbrave 150,[15] patients with oesophageal varices were excluded because there is a high risk of gastroesophageal varices-related haemorrhage in patients with hepatocellular carcinoma.[25] Patients who had an endoscopy within 3 months, with evidence of grade 3 oesophageal varices, were excluded from the study because of the invasive nature of endoscopy. In ORIENT-32, for patients who had not undergone previous endoscopy, endoscopy was done at the discretion of investigator to identify those with a high risk of bleeding based on clinical manifestation and CT scan of portal hypertension. Therefore, the population in our study might have been more representative of the population observed in routine clinical practice for management of hepatocellular carcinoma.

This study was done during the COVID-19 pandemic and was afftected by travel restrictions; regular treatment and evaluation visits were affected. To deal with these challenges, specific measures were taken, including delivering sorafenib to patients in accessible regions, assessing and treating patients at local hospitals or through teleconsultation (if patient is stable) and delaying treatment if appropriate. Delays in drug administration and tumour assessment might have delayed the timely confirmation of disease response, and therefore might have attenuated the actual objective response rate. The effect of COVID-19 on progression-free survival was unclear, since delayed drug administration might accelerate disease progression (shortening the actual progression-free survival). However, the delayed radiographic assessment of patients who progressed might extend the observation period for disease progression, and therefore prolong progression-free survival for these patients.

Although PD-L1 expression has been reported to be predictive of survival and efficacy benefit in several solid tumours, such as non-small-cell lung cancer and renal cell carcinoma,[26,27] no clear association has been reported for hepatocellular carcinoma.[15,28] Therefore, additional biomarker analyses using RNA sequencing have been done to explore effective biomarkers for patient selection, and the results will be reported in a subsequent publication. Updated overall survival results will also be reported.

There are several limitations of the study. The study was done in the Chinese population, and therefore, our findings might not be generalisable to other populations. Biomarker analysis for treatment efficacy was not included in this Article, but is ongoing.

In conclusion, sintilimab–bevacizumab biosimilar showed a significant overal survival and progression-free survival benefit compared with sorafenib in Chinese patients with previously untreated, HBV-associated hepatocelluar carcinoma with a tolerable and manageable safety profile. This combination regimen could provide a novel treatment option for this patient population.

Contributors

ZR and JF contributed to the study design. JF was the chief investigator. JX, YB, AX, SC, CD, QL, YL, YC, YG, ZC, BL, WJ, JianWu, JuWa, GS, BZ, YS, ZM, JianbingWu, SG, WY, CL, XS, ZG, TY, JC, MH, BX, YM, GT, YQ, JiWa, FX, GY and the ORIENT-32 study group contributed to patient enrolment and data acquisition. MC did the statistical analysis. YY, YW, and HZ contributed to the data interpretation and medical reviewing. ZR, YY, and YW drafted the manuscript. All authors contributed to manuscript revision. ZR, JF, and MC accessed and verified the raw data in the manuscript and had final responsibility for the decision to submit for publication.

Declaration of interests

ZR has received honoraria as an advisor from Innovent Biologics, AstraZeneca, F Hoffmann-La Roche, and Merck Sharp & Dohme. YY, MC, YW. and HZ are employees of Innovent Biologics. All other authors declare no competing interests.

Data sharing

The study protocol is available in appendix 2. Individual participant data that underlie the results reported in here, can be made available to other qualified medical investigators after de-identification, via a request to data@innoventbio.com. Innovent Biologics will not share data dictionaries or data from identified participants. Innovent Biologics will assess and approve the proposals at its own discretion. The requestors for access to the data will need to sign a data access agreement. Data will be available beginning 12 months and ending 36 months after Article publication.

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